

Three exposures. One world in color.

Reconstructing the Prokudin-Gorskii collection through channel alignment, exhaustive search and coarse-to-fine image pyramids.

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18 + 2

Provided + additional plates

6

High-resolution input scans

2.74 s

Slowest provided alignment



Handwritten NCC pyramid · original scan processed at 3741×9715

Recover the alignment, then recover the color.

The inputs are grayscale scans containing three vertically stacked exposures in blue, green and red order. The program divides each scan into three equal-height arrays, discards at most two trailing rows, keeps blue fixed and translates green and red. The final channel order is red, green, blue.

Stacked B / G / R → Split & normalize → Estimate G / R shifts → Compose RGB

Single-scale search

For each moving channel, try all 961 integer shifts in a ± 15 -pixel window. Compare a fixed interior region of blue against the shifted channel. Exclude a 12% scoring border on every side, expanding the exclusion when needed to keep every tested coordinate in bounds.

All candidates at a level use exactly the same reference pixels. This avoids favoring shifts that compare a smaller overlap or introduce wraparound.

Two baseline metrics

$$L2 = \sqrt{\sum (B - C_{\text{shifted}})^2}$$

Choose the smallest distance. L2 is sensitive to differences in brightness across color filters.

$$NCC = \Sigma(B \cdot C_{\text{shifted}}) / (\|B\| \|C_{\text{shifted}}\|)$$

Choose the largest similarity. This is the normalized dot product in the handout, without mean subtraction. It compensates for multiplicative intensity changes, but not arbitrary color-dependent changes.

Convert unsigned integer pixels to floating point using their dtype range. Use Pillow only for reading, writing and resizing; NumPy performs the manual search and array operations. No automatic registration or pyramid-building library is used.

Start small. Make each step visible.

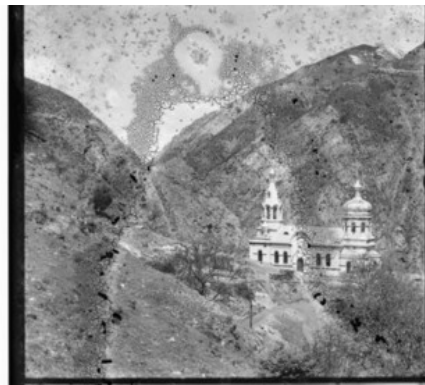
Every provided plate receives a low-resolution experiment using channels resized to 320 pixels wide. Both L2 and NCC run on the same pixels and search window. The three examples below show the separated exposures, unaligned composite and two aligned composites. Offsets here are in **low-resolution pixels**.

DETAILED EXPERIMENT 01

00056v.jpg



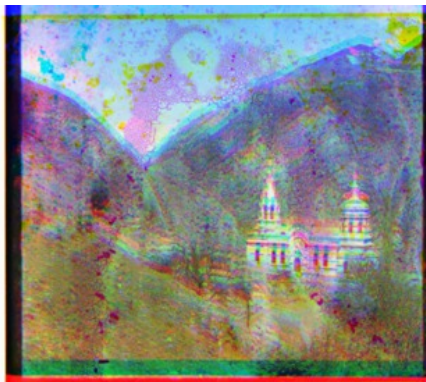
B channel



G channel



R channel



Unaligned RGB



Single-scale L2



Single-scale NCC

Low-resolution shifts: G (1, 4) · R (0, 10); same for L2 and NCC.

The church and hillside become much easier to read after registration. The colored speckles remain: the separate source exposures contain different scratches and blotches. A translation cannot restore missing emulsion.

DETAILED EXPERIMENT 02

00125v.jpg



B channel



G channel



R channel



Unaligned RGB



Single-scale L2



Single-scale NCC

Low-resolution shifts: $G(2, 4) \cdot R(1, 8)$; same for L2 and NCC.

The skyline, church and riverbank provide stable geometric structure. The two metrics select the same shifts at this reduced resolution. Narrow colored bands remain near the original plate edges.

00163v.jpg



B channel



G channel



R channel



Unaligned RGB



Single-scale L2



Single-scale NCC

Low-resolution shifts: G (1, -2) · R (1, -3); same for L2 and NCC.

Tree trunks and roof lines provide alignment cues. The negative vertical shifts are valid: these channels must move upward relative to blue. The two metrics again select the same low-resolution shifts.

All low-resolution experiments

Each shift cell lists G first, then R. Times are NCC / L2 in seconds. Plate numbers map to the labeled results below.

Plate	NCC shifts	L2 shifts	Time (s)	Images
Plate 01	(1, 4)	(1, 4)	0.383 / 0.237	NCC · L2
	(0, 10)	(0, 10)		
Plate 02	(2, 4)	(2, 4)	0.387 / 0.236	NCC · L2
	(1, 8)	(1, 8)		
Plate 03	(1, -2)	(1, -2)	0.378 / 0.256	NCC · L2
	(1, -3)	(1, -3)		
Plate 04	(0, 4)	(0, 4)	0.442 / 0.287	NCC · L2
	(3, 8)	(3, 8)		

Plate	NCC shifts	L2 shifts	Time (s)	Images
Plate 05	(-2, 5) (-4, 10)	(-2, 5) (-4, 10)	0.441 / 0.269	NCC · L2
Plate 06	(1, 5) (1, 11)	(1, 5) (1, 11)	0.382 / 0.259	NCC · L2
Plate 07	(2, 2) (3, 6)	(2, 2) (3, 6)	0.474 / 0.315	NCC · L2
Plate 08	(1, 5) (3, 9)	(1, 5) (2, 9)	0.416 / 0.281	NCC · L2
Plate 09	(2, 5) (2, 11)	(2, 5) (2, 11)	0.400 / 0.241	NCC · L2
Plate 10	(2, 5) (1, 10)	(1, 5) (1, 10)	0.409 / 0.244	NCC · L2
Plate 11	(1, 6) (1, 13)	(1, 6) (1, 13)	0.385 / 0.258	NCC · L2
Plate 12	(0, 7) (-1, 13)	(0, 7) (-1, 14)	0.391 / 0.249	NCC · L2
Plate 13	(1, 5) (1, 10)	(1, 5) (1, 10)	0.509 / 0.346	NCC · L2
Plate 14	(4, 7) (7, 14)	(4, 7) (6, 14)	0.451 / 0.267	NCC · L2
Plate 15	(1, 7) (1, 15)	(1, 7) (1, 15)	0.413 / 0.246	NCC · L2
Plate 16	(3, 6) (5, 13)	(3, 6) (5, 13)	0.404 / 0.256	NCC · L2
Plate 17	(2, 4) (2, 9)	(2, 4) (2, 9)	0.381 / 0.265	NCC · L2
Plate 18	(0, 7) (0, 11)	(0, 7) (0, 11)	0.389 / 0.244	NCC · L2

Large shifts. Small searches.

Explicitly build a list of progressively smaller images using bilinear downsampling by about two. Stop when the shorter channel dimension is at most 160 pixels. At the smallest level, search ± 15 pixels. At each finer level, multiply the estimate by the actual width/height ratios, round to integer pixels and refine within ± 2 pixels. Using actual ratios handles odd image sizes.

All 18 provided scans are processed at their original resolution. The same NCC metric and parameters apply to every image; no per-image tuning is used. A finer level tests only 25 candidates per moving channel, instead of exhaustively searching for shifts exceeding 100 pixels at full size.

A pyramid in motion

Intermediate estimates for 00458u.jpg; offsets are measured in pixels of the indicated level.

Channel size	G shift	R shift	Search radius
117 × 102	(0, 1)	(1, 3)	± 15
234 × 203	(0, 3)	(2, 6)	± 2
468 × 405	(1, 5)	(4, 11)	± 2
936 × 810	(1, 11)	(8, 22)	± 2
1871 × 1619	(3, 21)	(16, 44)	± 2
3741 × 3238	(6, 43)	(32, 87)	± 2

Output cropping removes only pixels without support in all three shifted channels. This is a geometric validity crop, **not** automatic detection of the original photographic borders; colored plate borders therefore remain visible.

Original-resolution offsets and runtime

Shifts are (x, y), positive right/down, applied to G and R relative to B. Timing covers both channel alignments, including pyramid construction, but excludes loading, compositing and image saving. Median: 0.089 s; range: 0.078–2.741 s on this Windows environment. No full-resolution brute-force timing comparison was performed.

Plate	G shift	R shift	Levels	Time (s)
Plate 01	(1, 6)	(1, 13)	3	0.078
Plate 02	(2, 5)	(1, 10)	3	0.081

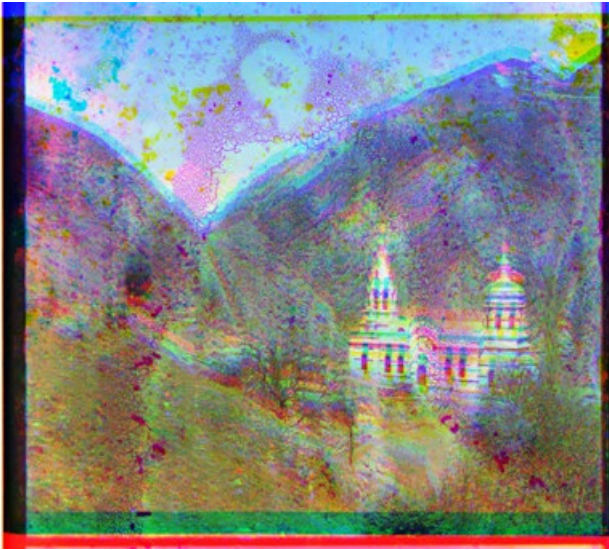
Plate	G shift	R shift	Levels	Time (s)
Plate 03	(1, -3)	(1, -4)	3	0.079
Plate 04	(6, 43)	(32, 87)	6	2.390
Plate 05	(-2, 6)	(-4, 13)	3	0.087
Plate 06	(14, 60)	(13, 129)	6	2.382
Plate 07	(20, 24)	(33, 71)	6	2.741
Plate 08	(2, 6)	(3, 11)	3	0.090
Plate 09	(2, 6)	(3, 13)	3	0.080
Plate 10	(2, 6)	(2, 13)	3	0.079
Plate 11	(1, 7)	(1, 16)	3	0.081
Plate 12	(0, 8)	(-1, 16)	3	0.096
Plate 13	(9, 55)	(12, 117)	6	2.473
Plate 14	(44, 77)	(77, 161)	6	2.417
Plate 15	(1, 8)	(1, 18)	3	0.091
Plate 16	(38, 71)	(63, 148)	6	2.459
Plate 17	(2, 6)	(3, 12)	3	0.086
Plate 18	(0, 8)	(0, 13)	3	0.086

04 / ALL PROVIDED IMAGES

The complete reconstruction set.

Each pair shows the unaligned composite and the NCC pyramid result. Click an aligned image to open the saved result at original pixel scale, after the common-support crop. Web previews are at most 900 pixels wide.

00056v.jpg



Before · unaligned RGB



After · NCC pyramid

G (1, 6) · R (1, 13) · applied at original resolution

Source damage remains visible as colored blotches.

PLATE 02 / PROVIDED

00125v.jpg

400 × 1024 input

0.081 s alignment



Before · unaligned RGB



After · NCC pyramid

G (2, 5) · R (1, 10) · applied at original resolution

PLATE 03 / PROVIDED

00163v.jpg

387 × 1024 input

0.079 s alignment



Before · unaligned RGB



After · NCC pyramid

G (1, -3) · R (1, -4) · applied at original resolution

PLATE 04 / PROVIDED

3741 × 9715 input

2.390 s alignment

00458u.jpg



Before · unaligned RGB



After · NCC pyramid

G (6, 43) · R (32, 87) · applied at original resolution

PLATE 05 / PROVIDED

396 × 1024 input

0.087 s alignment

00804v.jpg



Before · unaligned RGB



After · NCC pyramid

G (-2, 6) · R (-4, 13) · applied at original resolution

PLATE 06 / PROVIDED

01007a.jpg

3800 × 9715 input

2.382 s alignment



Before · unaligned RGB

G (14, 60) · R (13, 129) · applied at original resolution



After · NCC pyramid

PLATE 07 / PROVIDED

01047u.jpg

3741 × 9656 input

2.741 s alignment



Before · unaligned RGB

G (20, 24) · R (33, 71) · applied at original resolution



After · NCC pyramid

PLATE 08 / PROVIDED

01164v.jpg

396 × 1024 input

0.090 s alignment



Before · unaligned RGB



After · NCC pyramid

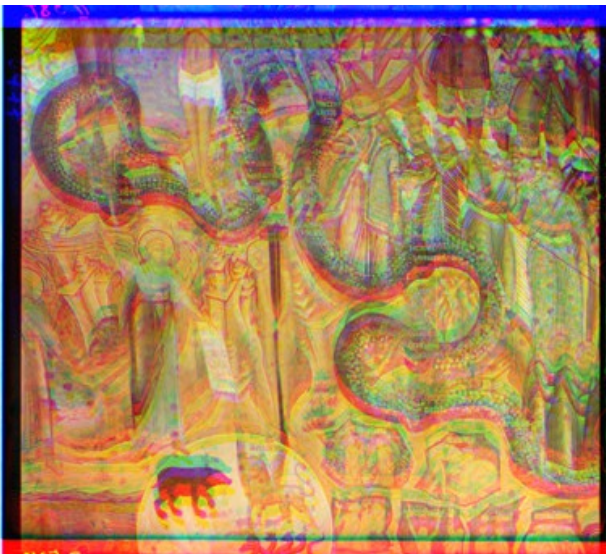
G (2, 6) · R (3, 11) · applied at original resolution

PLATE 09 / PROVIDED

01269v.jpg

395 × 1024 input

0.080 s alignment



Before · unaligned RGB



After · NCC pyramid

G (2, 6) · R (3, 13) · applied at original resolution

01522v.jpg



Before · unaligned RGB



After · NCC pyramid

G (2, 6) · R (2, 13) · applied at original resolution

Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

01597v.jpg



Before · unaligned RGB



After · NCC pyramid

G (1, 7) · R (1, 16) · applied at original resolution

Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

01598v.jpg



Before · unaligned RGB



After · NCC pyramid

G (0, 8) · R (-1, 16) · applied at original resolution

Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

01657u.jpg



Before · unaligned RGB



After · NCC pyramid

G (9, 55) · R (12, 117) · applied at original resolution

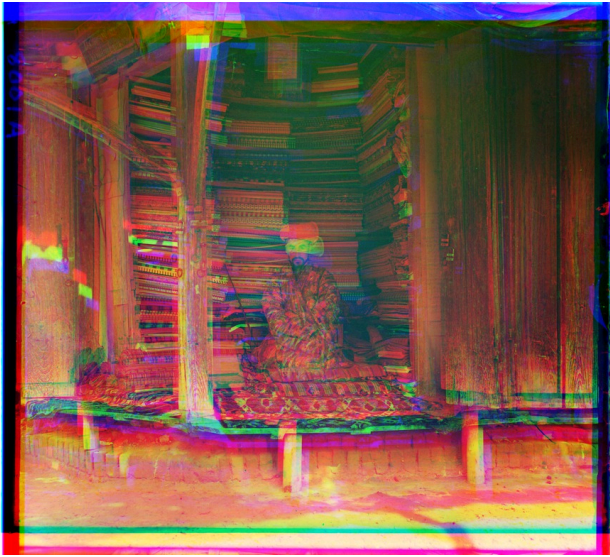
Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

PLATE 14 / PROVIDED

3731 × 9695 input

2.417 s alignment

01725u.jpg



Before · unaligned RGB



After · NCC pyramid

G (44, 77) · R (77, 161) · applied at original resolution

PLATE 15 / PROVIDED

397 × 1024 input

0.091 s alignment

01728v.jpg



Before · unaligned RGB



After · NCC pyramid

G (1, 8) · R (1, 18) · applied at original resolution

01861a.jpg



Before · unaligned RGB



After · NCC pyramid

G (38, 71) · R (63, 148) · applied at original resolution

Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

PLATE 17 / PROVIDED

10131v.jpg

400 × 1024 input

0.086 s alignment



Before · unaligned RGB



After · NCC pyramid

G (2, 6) · R (3, 12) · applied at original resolution

PLATE 18 / PROVIDED

31421v.jpg

398 × 1024 input

0.086 s alignment



Before · unaligned RGB



After · NCC pyramid

G (0, 8) · R (0, 13) · applied at original resolution

Beyond the supplied dataset.

Two additional three-frame grayscale JPEGs were downloaded from the Library of Congress, rather than using its finished color restorations. Both use exactly the same pipeline and parameters as the provided set. These examples are low-resolution scans; the large-image demonstration comes from the six supplied high-resolution scans.

ADDITIONAL COLLECTION EXAMPLE

00166v.jpg

V Malorossii (In Little Russia [i.e. Ukraine])



Before · unaligned RGB



After · NCC pyramid

G (2, 3) · R (4, 12) · 0.085 s

Official catalog record · Original grayscale scan · Single-scale NCC · Single-scale L2

ADDITIONAL COLLECTION EXAMPLE

00637v.jpg

Probnaiia shtolnia (Trial adit; iron ore piled in front)



Before · unaligned RGB



After · NCC pyramid

G (-2, 4) · **R** (-5, 10) · 0.094 s

Official catalog record · Original grayscale scan · Single-scale NCC · Single-scale L2

06 / DISCUSSION & LIMITATIONS

Alignment is only part of restoration.

Visual assessment: the reconstructed previews show broadly aligned scene structure. This is a qualitative inspection, not a ground-truth accuracy score. The saved high-resolution outputs remain available for closer inspection.

Damaged exposures: the first detailed experiment has visible scratches and blotches in its individual grayscale channels. Their disagreement produces colored speckles after composition. Searching for a different global displacement cannot reconstruct these missing details.

Local residuals: faces, foliage and object boundaries can retain colored fringes. Motion between sequential exposures, channel-dependent brightness and departures from pure translation are plausible causes; the current experiment does not isolate their individual contributions. The model cannot correct local motion, rotation or scale change.

Border artifacts: excluding borders from the score prevents them dominating the alignment, but does not restore or remove the physical borders in the final image. The common-support crop only removes wrapped pixels.

Optional extensions: edge-based alignment, detected-border cropping, white balance and contrast enhancement are not implemented in this baseline submission. Keeping the original intensities makes the L2/NCC results and remaining defects transparent.

Dataset discrepancy: the handout mentions both 5 and 20 examples and legacy image names; the supplied directory contains 18 numbered JPEGs. All 18 were processed, and three selected plates receive detailed experiments.

07 / REPRODUCIBILITY

Measurements, code & sources.

Seven synthetic-image unit tests cover channel order, 16-bit normalization, L2/NCC shift recovery, large displacements on odd dimensions, RGB composition, valid-support cropping, blank-image tie handling and invalid inputs. No mock tests are used.

Run instructions are in the project README. Saved records include input SHA-256 hashes, channel dimensions, discarded rows, offsets, per-level scores and timings. NCC/L2 scores at different pyramid levels use different pixels and should not be interpreted as one comparable accuracy curve.

Environment: Python 3.13.5, NumPy 2.3.3, Pillow 11.1.0. Measurements were collected with one sequential batch, not a statistical performance benchmark.

Source images: Prokudin-Gorskii photograph collection, Library of Congress, Prints and Photographs Division. See the linked official records above and the collection overview. Assignment requirements: local SYDE671 Assignment 1 handout, Part Two.